Publication List

Peter B. Denton

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Articles (79)

[1] P. B. Denton, S.-F. Ge, C.-F. Kong, and P. Pasquini, "RG Running of Multiple Neutrino Mixing Parameters at Oscillation Experiments," arXiv:2606.26720 [hep-ph].

[2] P. B. Denton, J. Gehrlein, and H. Truelson, "The Future of Lepton Flavor," arXiv:2606.05291 [hep-ph].

[3] J. Banovetz, C.-A. Hebert, P. B. Denton, D. Scolnic, A. Slosar, and C. Walter, "Uncovering the Next Galactic Supernova with the Vera C. Rubin Observatory," *The Open Journal of Astrophysics* (1, 2026) , arXiv:2601.12094 [astro-ph.IM].

*For the latest version see: peterdenton.github.io

†Most author lists are in alphabetical order as that is the standard in particle physics.

[4] P. B. Denton and S. J. Parke, "Efficient atmospheric, solar, and supernova neutrino propagation through the Earth," *Phys. Rev. D* **113** no. 5, (2026) 053002, arXiv:2511.04735 [hep-ph].

[5] L. A. Anchordoqui *et al.*, "Letter of Intent: The Forward Physics Facility," arXiv:2510.26260 [hep-ex].

[6] H. Davoudiasl, P. B. Denton, and A. M. Suliga, "Glimpsing Physics of Nano-Hz Gravitational Waves in Neutrinos from Core-Collapse Supernovae," arXiv:2510.23713 [hep-ph].

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[10] P. B. Denton, "Neutrino Oscillations in the Three Flavor Paradigm," *Physics* **3** (2026) 194, arXiv:2501.08374 [hep-ph].

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Conference Proceedings

[1] **GRAND** Collaboration, R. Alves Batista *et al.*, "The Giant Radio Array for Neutrino Detection (GRAND) Collaboration – Contributions to the 38th International Cosmic Ray Conference (ICRC 2023)," in *38th International Cosmic Ray Conference*. 7, 2023. arXiv:2308.00120 [hep-ex].

[2] **UHECR** Collaboration, F. Schroeder *et al.*, "Snowmass UHECR Whitepaper: Requirements on Future Instrumentation," *PoS* **ICRC2023** (2023) 206.

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Collaboration Papers

[1] **DUNE** Collaboration, S. Abbaslu *et al.*, "A Bayesian approach to the long-baseline neutrino oscillation sensitivity of DUNE," arXiv:2608.04059 [hep-ex].

[2] **DUNE** Collaboration, S. Abbaslu *et al.*, "Charge readout electronics for the DUNE horizontal drift far detector: design and performance in ProtoDUNE-HD," *JINST* **21** (2026) P08018, arXiv:2604.23966 [physics.ins-det].

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[4] **DUNE** Collaboration, S. Abbaslu *et al.*, "Measurement of Exclusive π^+–argon Interactions Using ProtoDUNE-SP," arXiv:2511.13462 [hep-ex].

[5] **DUNE** Collaboration, S. Abbaslu *et al.*, "First Measurement of π^+-Ar and p-Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data," arXiv:2511.11925 [nucl-ex].

[6] **DUNE** Collaboration, S. Abbaslu *et al.*, "Identification of low-energy kaons in the ProtoDUNE-SP detector," arXiv:2510.08380 [hep-ex].

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[10] **DUNE** Collaboration, S. Abbaslu *et al.*, "Spatial and Temporal Evaluations of the Liquid Argon Purity in ProtoDUNE-SP," arXiv:2507.08586 [physics.ins-det].

[11] **DUNE** Collaboration, A. Abed Abud *et al.*, "European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment," arXiv:2503.23744 [physics.acc-ph].

[12] **DUNE** Collaboration, A. Abed Abud *et al.*, "DUNE Software and Computing Research and Development," arXiv:2503.23743 [physics.data-an].

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[14] **DUNE** Collaboration, A. Abed Abud *et al.*, "The DUNE Phase II Detectors," arXiv:2503.23293 [physics.ins-det].

[15] **DUNE** Collaboration, A. Abed Abud *et al.*, "Neutrino interaction vertex reconstruction in DUNE with Pandora deep learning," *Eur. Phys. J. C* **85** no. 697, (2025) 697, arXiv:2502.06637 [hep-ex].

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[17] **DUNE** Collaboration, A. Abed Abud *et al.*, "DUNE Phase II: scientific opportunities, detector concepts, technological solutions," *JINST* **19** no. 12, (2024) P12005, arXiv:2408.12725 [physics.ins-det].

[18] **GRAND** Collaboration, R. Alves Batista *et al.*, "GRANDlib: A simulation pipeline for the Giant Radio Array for Neutrino Detection (GRAND)," *Comput. Phys. Commun.* **308** (2025) 109461, arXiv:2408.10926 [astro-ph.IM].

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[21] **DUNE** Collaboration, A. Abed Abud *et al.*, "Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber," *Instruments* **8** no. 3, (2024) 41, arXiv:2403.03212 [physics.ins-det].

[22] **DUNE** Collaboration, A. Abed Abud *et al.*, "Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light," *JINST* **19** no. 08, (2024) P08005, arXiv:2402.01568 [physics.ins-det].

[23] **DUNE** Collaboration, A. Abed Abud *et al.*, "The DUNE Far Detector Vertical Drift Technology. Technical Design Report," *JINST* **19** no. 08, (2024) T08004, arXiv:2312.03130 [hep-ex].

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[25] **DUNE** Collaboration, A. Abed Abud *et al.*, "Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment," *Phys. Rev. D* **107** no. 11, (2023) 112012, arXiv:2303.17007 [hep-ex].

[26] **DUNE** Collaboration, A. Abed Abud *et al.*, "Highly-parallelized simulation of a pixelated LArTPC on a GPU," *JINST* **18** no. 04, (2023) P04034, arXiv:2212.09807 [physics.comp-ph].

[27] **FASER** Collaboration, H. Abreu *et al.*, "Technical Proposal: FASERnu," arXiv:2001.03073 [physics.ins-det].

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[29] **GRAND** Collaboration, J. Álvarez Muñiz *et al.*, "The Giant Radio Array for Neutrino Detection (GRAND): Science and Design," *Sci. China Phys. Mech. Astron.* **63** no. 1, (2020) 219501, arXiv:1810.09994 [astro-ph.HE].

Talks (132 including 90 invited)

[1] "Astrophysical Neutrino Decay at IceCube.". **Invited** talk at Stony Brook University, October 2026.

[2] "The Future of Lepton Flavor." https://indico.bnl.gov/event/33058/. Talk at the BNL HET Group, July 2026.

[3] "CP Violation from CP Conservation.". **Invited** chalk talk at Fermilab theory group, June 2026.

[4] "Nu Theory.". **Invited** colloquium at University of Tennessee, April 2026.

[5] "CP-Violation with Neutrino Disappearance and NuFast." https://indico.bnl.gov/event/31076/. **Invited** seminar at Kent State University, April 2026.

[6] "Nu Theory." https://indico.bnl.gov/event/31320/. **Invited** colloquium at BNL, March 2026.

[7] "NuFast." https://indico.bnl.gov/event/31076/. Talk at the BNL HET Group, February 2026.

[8] "CP-Violation with Neutrino Disappearance and NuFast.". **Invited** seminar at Oxford, UK, January 2026.

[9] "CP-Violation with Neutrino Disappearance and NuFast.". **Invited** seminar at Imperial College London, January 2026.

[10] "CP-Violation with Neutrino Disappearance and NuFast.". **Invited** seminar at Kings College London, January 2026.

[11] "Modern Neutrino Oscillation Theory."
https://indico.cern.ch/event/1568942/contributions/6741263/. **Invited** plenary talk at NuPhys, Kings College London, January 2026.

[12] "NuFast."
https://indico.phys.sinica.edu.tw/event/301/contributions/2540/. **Invited** plenary talk at the International Joint Workshop on the Standard Model and Beyond, Taipei, December 2025.

[13] "Neutrino Oscillations in the Three Flavor Paradigm."
youtube.com/watch?v=OxjUAne79M4.
https://neutrinos.discussingresearch.com/. **Invited** pedagogical plenary for Neutrinos From Home, December 2025.

[14] "NuFast.". **Invited** talk at Stony Brook, November 2025.

[15] "CP-Violation with Neutrino Disappearance and NuFast.". **Invited** seminar at NYU, November 2025.

[16] "Solar Neutrinos and the Strongest Oscillation Constraints on Scalar NSI.". **Invited** talk, August 2025.

[17] "Untangling Systematic Uncertainties from BSM Physics."
https://indico.cern.ch/event/1424413/contributions/6468090/. **Invited** plenary at NPN at University of Cincinnati, June 2025.

[18] "Modern Neutrino Oscillation Theory."
https://agenda.hep.wisc.edu/event/2257/contributions/33831/. **Invited** plenary at CIPAN at University of Wisconsin, Madison, June 2025.

[19] "Towards Solidifying the Discrete Mass Ordering Measurement in the Presence of New Physics."
https://indico.cern.ch/event/1458241/contributions/6507839/. **Invited** plenary at Magnificent CEvNS, Rio, June 2025.

[20] "NuFast.". **Invited** talk at the Stony Brook DUNE group, June 2025.

[21] "Shining the Neutrino Flashlight into the Darkest Places.". **Invited** seminar, June 2025.

[22] "Testing the Three-Flavor Paradigm."
https://indico.global/event/13925/contributions/125625/. **Invited** plenary at the NEAT: Neutrino Experiment and Theory workshop at Colorado State University, May 2025.

[23] "Individual Neutrino Masses From a Supernova."
https://indico.cern.ch/event/1488822/contributions/6480082/. **Invited** plenary at the Mitchell Conference at Texas A&M, May 2025.

[24] "Neutrino Factory.". **Invited** seminar at Universidad Catolica del Norte, April 2024.

[25] "Neutrino Factory." https://indico.bnl.gov/event/26010/. Talk at the BNL HET Group, March 2025.

[26] "CP Violation with Neutrino Disappearance." https://indico.cern.ch/event/1454726/contributions/6376522/. Talk at the CERN Neutrino Platform Pheno Week, February 2025.

[27] "Neutrino Oscillations: Unveiling Mysteries.". **Invited** colloquium at Virginia Tech, February 2025.

[28] "Individual Neutrino Masses From a Supernova." https://indico.bnl.gov/event/25821/. Talk at the BNL HET Group, January 2024.

[29] "Connecting the Extremes: A Story of Supermassive Black Holes and Ultralight Dark Matter." https://physics.ucsd.edu/events/seminars-colloquia/event?event_id=1396. **Invited** seminar at UCSD, October 2024.

[30] "Dark Matter Raining on DUNE and Other Large Volume Detectors." https://indico.bnl.gov/event/24793/. Talk at the BNL HET Group, October 2024.

[31] "Unitarity Violation in Neutrino Physics: Brief Pedagogy." https://indico.fnal.gov/event/66471/. **Invited** talk at DUNE collaboration call, October 2024.

[32] "Modern Neutrino Oscillation Theory." https://indico.fnal.gov/event/63406/contributions/297152/. **Invited** keynote plenary at NuFact, Argonne, September 2024.

[33] "NuFast: Fast and Accurate Algorithm for Calculating Long-Baseline Neutrino Oscillation Probabilities with Matter Effects." https://agenda.infn.it/event/39753/contributions/240076/. **Invited** talk at NOW, Otranto Italy, August 2024.

[34] "CP-Violation with Neutrino Disappearance and NuFast." https://indico.sanfordlab.org/event/69/timetable/#132-talk-dark-matter-raining-o. **Invited** talk at the Center for Center for Theoretical Underground Physics and Related Areas (CETUP*) workshop, Lead SD, July 2023.

[35] "NuFast: Fast and Accurate Algorithm for Calculating Long-Baseline Neutrino Oscillation Probabilities with Matter Effects." https://indico.fnal.gov/event/60082/contributions/291412/. Talk at DUNE Collaboration Meeting at Fermilab, May 2024.

[36] "CP Violation with Neutrino Disappearance.". **Invited** seminar at Rice University, April 2024.

[37] "CP Violation with Neutrino Disappearance.". **Invited** seminar at Colorado State University, March 2024.

[38] "CP Violation with Neutrino Disappearance." https://indico.bnl.gov/event/21760/. Talk at the BNL HET Group, March 2024.

[39] "CP Violation with Neutrino Disappearance." https://indico.fnal.gov/event/63560/. Talk at the DUNE LBL WG, March 2024.

[40] "Knowns and Unknowns in Neutrinos." https://web.mit.edu/lns/news/archives/index.html. **Invited** colloquium at MIT, Boston, October 2023.

[41] "LMA-Dark: Large New Physics Effects in Neutrino Oscillations.". **Invited** talk at the COHERENT collaboration, August 2023.

[42] "Testing Unitarity of the Leptonic Mixing Matrix with Oscillations: A Focus on Tau Neutrinos." https://ifirse.icise.vn/nugroup/nuworkshop2023/program.html. **Invited** plenary at the 19th Rencontres du Vietnam, July 2023.

[43] "Light Sterile Neutrinos: A Modern Picture and a Model to Evade Cosmology." https://indico.ihep.ac.cn/event/18269/contributions/135575/. **Invited** talk at WIN, July 2023.

[44] "Light Sterile Neutrinos: A Modern Picture and a Model to Evade Cosmology." https://indico.sanfordlab.org/event/53/contributions/822/. **Invited** talk at the Center for Center for Theoretical Underground Physics and Related Areas (CETUP*) workshop, Lead SD, July 2023.

[45] "Here Comes the Sun: Solar Parameters in Long-Baseline Accelerator Neutrino Oscillations." https://indico.cern.ch/event/1218225/contributions/5384272/. Talk at Pheno, Pittsburgh, May 2023.

[46] "Light Sterile Neutrinos: A Modern Picture and a Model to Evade Cosmology." https://indico.cern.ch/event/1258338/contributions/5307365/. **Invited** talk at University of Cape Town, April 2023.

[47] "Knowns and Unknowns in Neutrinos.". **Invited** colloquium at University of Wisconsin, Madison, April 2023.

[48] "Neutrinos at Snowmass.". **Invited** seminar at Kings College London, January 2023.

[49] "Knowns and Unknowns in Neutrinos.". **Invited** colloquium at Stony Brook University, New York, October 2022.

[50] "Light (Fermionic?) Dark Matter." https://indico.cern.ch/event/1189979/contributions/5012521/. **Invited** talk at the International Conference on Neutrinos and Dark Matter in Egypt, October 2022.

[51] "Connecting the Extremes: A Story of Supermassive Black Holes and Ultralight Dark Matter.". **Invited** talk at Dark Matter in Compact Objects, Stars, and in Low Energy Experiments at INT, Seattle, August 2022.

[52] "Tau Neutrinos: from GeV to EeV." https://indico.fnal.gov/event/22303/contributions/246362/. **Invited** talk at Snowmass, Seattle, July 2022.

[53] "Sterile neutrinos at 1 eV." https://www.mpi-hd.mpg.de/lin/seminar_theory.en.php. **Invited** seminar at MPI Heidelberg, July 2022.

[54] "Connecting the Extremes: Story of Supermassive Black Holes and Ultralight Dark Matter." https://n3as.berkeley.edu/p/event/su22-jun14/. **Invited** seminar at UC Berkeley via N3AS, June 2022.

[55] "Connecting the Extremes: Story of Supermassive Black Holes and Ultralight Dark Matter." https://theory.tifr.res.in/~sotu/previous.php. **Invited** seminar at Tata Institute of Fundamental Research, India, June 2022.

[56] "Flavor mixing, CP violation, and Unitarity." https://neutrino2022.org/program/detail_program. **Invited** plenary talk at Neutrino 2022, Seoul Korea, June 2022.

[57] "CP Violation at Long-Baseline Neutrino Experiments." https://indico.cern.ch/event/1125426/contributions/4868720/. **Invited** plenary talk at the Mitchell Conference on Collider, Dark Matter, and Neutrino Physics; Texas A&M, College Station TX, May 2022.

[58] "Neutrino Theory Overview." https://indico.sanfordlab.org/event/28/contributions/310/. **Invited** plenary talk at the Conference on Science at the Sanford Underground Research Facility (CoSSURF), May 2022.

[59] "Nu physics: Theory and practice." https://indico.cern.ch/event/1089132/contributions/4863585/. **Invited** plenary talk at the Phenomenology Symposium, Pittsburgh, May 2022.

[60] "CP Violation at Long-Baseline Neutrino Experiments." https://lawphysics.wordpress.com/2022/04/20/w131-peter-denton-cp-violation-at-long- **Invited** seminar in the Latin American Webinar Physics series, April 2022.

[61] "CP Violation at Long-Baseline Neutrino Experiments.". **Invited** seminar at Harvard, April 2022.

[62] "Connecting the Extremes: Story of Supermassive Black Holes and Ultralight Dark Matter.". **Invited** seminar at MIT, April 2022.

[63] "New Perspectives on Atmospheric Neutrinos.". **Invited** seminar at INFN Torino, March 2022.

[64] "Tau Neutrino Identification at IceCube for Unitary Violation Tests." https://indico.cern.ch/event/1103445/contributions/4724172/. Talk at Snowmass BSM neutrino workshop, February 2022.

[65] "Neutrinos and Cosmic Rays at Snowmass." https://indico.bnl.gov/event/13887/. Talk at BNL Snowmass Retreat, December 2021.

[66] "Astrophysical Neutrino Decay." https://indico.ipmu.jp/event/397/contributions/6390/. **Invited** talk at Dark Sectors of Astroparticle Physics at IPMU, Japan December 2021.

[67] "Neutrino Oscillations at FPF." https://indico.cern.ch/event/1076733/contributions/4577119/. Talk at Third Forward Physics Facility October 2021.

[68] "Astrophysical Neutrino Decay." https://indico.ific.uv.es/event/6178/contributions/15526/. Talk at TAUP August 2021.

[69] "Astrophysical Neutrino Decay." https://indico.desy.de/event/28202/contributions/105961/. Talk at EPS-HEP July 2021.

[70] "CP-Violating Neutrino Non-Standard Interactions in Long-Baseline-Accelerator Data." https://indico.cern.ch/event/1034469/contributions/4430079/. Talk at DPF at FSU July 2021.

[71] "CP-Violating Neutrino Non-Standard Interactions in Long-Baseline-Accelerator Data." https://indico.ibs.re.kr/event/357/timetable/. Talk at PASCOS in IBS Koreas June 2021.

[72] "CP-Violating Neutrino Non-Standard Interactions in Long-Baseline-Accelerator Data." https://indico.cern.ch/event/982783/contributions/4362341/. Talk at Pheno May 2021.

[73] "Neutrino Oscillations in Matter and Linear Algebra.". **Invited** colloquium at Illinois Institute of Technology April 2021.

[74] "CP Violation at Long-Baseline Neutrino Experiments.". **Invited** seminar at Michigan State University March 2021.

[75] "Astrophysical Neutrino Decay." https://agenda.infn.it/event/24250/contributions/129755/. Talk at the XIX International Workshop on Neutrino Telescopes February 2021.

[76] "CP Violation at Long-Baseline Neutrino Experiments." https://indico.cern.ch/event/1001277/. **Invited** seminar at Sydney CPPC February 2021.

[77] "Ultralight Fermionic Dark Matter.". **Invited** talk at Asymptotic Safety and Dark Matter workshop at OSU December 2020.

[78] "Ultralight Fermionic Dark Matter." https://www.ictp-saifr.org/dmw2020/. Talk at 3rd South American Dark Matter Workshop at ICTP in Sao Paulo December 2020.

[79] "3+1+NSI and CP Violation.". **Invited** seminar at KIAS November 2020.

[80] "CP Violation at Long-Baseline Neutrino Experiments." https://indico.bnl.gov/event/8008/. Talk at BNL HET Group October 2020.

[81] "CP Violation at Long-Baseline Neutrino Experiments." https://npc.fnal.gov/neutrino-seminar-series/. **Invited** Neutrino Physics Center seminar October 2020 at Fermilab.

[82] "The Lightest Dark Matter.". **Invited** seminar October 2020 at University of Sussex.

[83] "Astrophysical Neutrino Decay." https://indico.cern.ch/event/868940/contributions/3899680/. Talk at ICHEP July 2020 in Prague (virtual).

[84] "Visible Decay of Astrophysical Neutrinos." https://indico.bnl.gov/event/7985/. Talk at BNL HET Group May 2020.

[85] "Ultralight Boson Dark Matter Constraints from Superradiance Leveraging the Event Horizon Telescope Collaboration's Observations of M87*." https://indico.cern.ch/event/858682/contributions/3837326/. Talk at Pheno May 2020 in Pittsburgh, PA (virtual).

[86] "Beyond the Standard Model physics with accelerator neutrino experiments." https://aps-april.onlineeventpro.freeman.com/sessions/15336169/subsession/25117238/ **Invited** plenary at APS April Meeting 2020 (virtual).

[87] "LMA-Dark: Large New Physics Effects in Neutrino Oscillations." https://indico.bnl.gov/event/7665/. Talk at BNL HET Group February 2020.

[88] "Motivation for neutrino precision in oscillations." https://indico.bnl.gov/event/7282/. **Invited** talk at BNL Snowmass Intensity Frontier & Astrophysics Workshop February 2020.

[89] "Recent results in neutrino oscillation theory." https://www.physics.umass.edu/events/2019-11-15-recent-results-neutrino-oscillatior **Invited** seminar at UMass Amherst November 2019.

[90] "Realizing the physics goals at DUNE." https://indico.fnal.gov/event/21535/other-view. **Invited** talk at Modules Of Opportunity for DUNE workshop at BNL November 2019.

[91] "Recent results in neutrino oscillation theory." https://physics.osu.edu/events/high-energy-physics-seminar-peter-dentonbrookahaven- **Invited** seminar at OSU November 2019.

[92] "New physics probes in future neutrino experiments." https://indico.bnl.gov/event/6652/. **Invited** colloquium at BNL October 2019.

[93] "Recent results in neutrino oscillation theory." https://indico.cern.ch/event/800930/contributions/3557081/. Talk at CERN Neutrino Platform October 2019.

[94] "Neutrino theory in the coming years." https://indico.bnl.gov/event/6710/. **Invited** talk at BNL Snowmass Discussion October 2019.

[95] "Recent results in neutrino oscillation theory." https://theory.fnal.gov/events/event/tbd-neutrinos/. **Invited** theory seminar at Fermilab September 2019.

[96] "Exact neutrino oscillation probabilities in matter." https://indico.ific.uv.es/event/3649/contributions/11349/. Talk given at TomFest at Vanderbilt August 2019.

[97] "Neutrino oscillation probabilities in matter." https://indico.cern.ch/event/782953/contributions/3444777/. Talk given at the 2019 DPF meeting at Northeastern July 2019.

[98] "Neutrino self interactions in the early universe." https://indico.cern.ch/event/812851/contributions/3432032/. **Invited** talk at NTN NSI Workshop at Wash U May 2019.

[99] "Partial neutrino decay resolves icecube's track and cascade tension." https://indico.bnl.gov/event/5875/. Talk at BNL HET Group May 2019.

[100] "Neutrino Oscillation Probabilities in Matter." http://theory.physics.uci.edu/seminars.html. **Invited** seminar at UC Irvine May 2019.

[101] "Neutrino Oscillation Probabilities in Matter." http://www.theory.caltech.edu/people/carol/seminar.html. Seminar at Caltech May 2019.

[102] "Partial Neutrino Decay Addresses the Track – Cascade Tension at IceCube." https://indico.cern.ch/event/777988/contributions/3410555/. Talk at Pheno May 2019 in Pittsburgh, PA.

[103] "Neutrino Oscillation Probabilities in Matter." https://www.phys.psu.edu/seminars/all-seminars. **Invited** seminar at Penn State April 2019.

[104] "Neutrino Oscillation Probabilities in Matter." https://www.phys.vt.edu/Talks/NeutrinoPhysicsSeminar.html. **Invited** seminar at Virginia Tech February 2019.

[105] "Analytic and Compact Expressions for Neutrino Oscillations in Matter." https://dx.doi.org/10.5281/zenodo.2642372. **Invited** talk at PONDD workshop at Fermilab December 2018.

[106] "Finding the Unexpected in IceCube.". **Invited** N-Talk at Niels Bohr International Academy September 2018 in Copenhagen.

[107] "High Energy Neutrino Parameter Estimation.". **Invited** talk at GRAND workshop at IAP August 2018.

[108] "New Neutrino Interactions: Breaking Degeneracies and Relaxing Sterile Tensions.". **Invited** seminar at BNL August 2018.

[109] "Analytic and compact perturbative expressions for neutrino oscillations in matter." https://indico.cern.ch/event/686555/contributions/2977525/. Talk at the International Conference of High Energy Physics (ICHEP) July 2018 in Seoul.

[110] "Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.". **Invited** talk at IIHE April 2018 in Brussels.

[111] "Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.". **Invited** talk at DESY January 2018 in Zeuthen.

[112] "Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.". **Invited** talk at Arizona State University January 2018.

[113] "Supernova - Gamma Ray Burst - Neutrino Connection.". **Invited** SUPER-STARS talk at DARK Cosmology Center November 2017 in Copenhagen.

[114] "Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.". **Invited** N-Talk at Niels Bohr International Academy November 2017 in Copenhagen.

[115] "Analytic and compact perturbative expressions for neutrino oscillations in matter.". **Invited** seminar at Campinas State University October 2017.

[116] "COHERENT and the LMA-Dark NSI Solution." https://indico.uu.se/event/324/session/20/contribution/182. **Invited** talk at the NUFACT 2017 workshop September 2017 in Uppsala.

[117] "What We Can Tell About the Sources of IceCube's Neutrinos, and What IceCube Can Tell Us About Gamma Ray Bursts." http://astro.fnal.gov/events/event/tbd-35/. Astrophysics theory seminar at Fermilab August 2017 in Batavia, IL.

[118] "The Galactic Contribution to IceCube's Astrophysical Neutrino Flux." https://indico.cern.ch/event/615891/contributions/2608935/. Talk at TeV Particle Astrophysics at CCAPP in Columbus, OH.

[119] "Finding Anisotropies in Cosmic Rays and Neutrinos." http://nbia.nbi.ku.dk/nbia-seminars/nbia-seminar-peter-denton/. **Invited** seminar at the Niels Bohr International Academy astroparticle seminar April 2017 in Copenhagen.

[120] "Analytic and compact perturbative expressions for neutrino oscillations in matter.". Talk at the Center of Excellence for Particle Physics at the Terascale at the University of Melbourne December 2016.

[121] "Spherical Harmonics as a Tool for Finding Anisotropies in UHECR and Astrophysical Neutrino Fluxes.". **Invited** talk at the Danish Astroparticle Physics Meeting October 2016 in Odense.

[122] "The Standard Neutrino Oscillation Parameters and a Surprising Alternative Solution.". **Invited** N-Talk at Niels Bohr International Academy September 2016 in Copenhagen.

[123] "Analytic and compact perturbative expressions for neutrino oscillations in matter." http://indico.cern.ch/event/432527/contributions/1071859/. Talk at the International Conference of High Energy Physics (ICHEP) August 2016 in Chicago, IL.

[124] "Analytic and compact perturbative expressions for neutrino oscillations in matter." http://theory.fnal.gov/seminars/seminars.html. **Invited** talk at the Fermilab theory seminar July 2016 in Batavia, IL.

[125] "Methods for Probing New Physics at High Energies." https://events.vanderbilt.edu/index.php?eID=90084. Successful dissertation defense at Vanderbilt University June 2016 in Nashville, TN.

[126] "Analytic and compact perturbative expressions for neutrino oscillations in matter." http://www.ccsem.infn.it/issp2016/index.html. Talk at the International School of Subnuclear Physics May 2016 in Erice, Sicily.

[127] "Analytic and compact perturbative expressions for neutrino oscillations in matter." https://indico.cern.ch/event/489180/contributions/2158195/. Talk at Pheno May 2016 in Pittsburgh, PA.

[128] "Cosmic Ray Anisotropy with Partial Sky Exposure.". **Invited** seminar November 2015 at CCAPP.

[129] "The Effect of a Maximum Lepton Energy on the Stability of Pions and Cosmic Ray Physics." http://meetings.aps.org/link/BAPS.2015.APR.M14.1. Talk at the APS April meeting 2015 in Baltimore, MD.

[130] "Particle Physics at the Highest Energies.". **Invited** seminar December 2014 at the University of Wisconsin – Madison.

[131] "Sensitivity of orbiting JEM-EUSO to large-scale cosmic-ray anisotropies.". Talk at the Cosmic Ray Anisotropy Workshop September 2013 in Madison, WI.

[132] "Using dispersion relations to look for new physics in pp elastic scattering at the LHC." http://meetings.aps.org/link/BAPS.2013.APR.H12.8. Talk at the APS April meeting 2013 in Denver, CO.

Lectures

[1] "Neutrino Theory Overview." https://indico.global/event/16507/overview. Lecture at the International Neutrino Summer School at UC Santa Barbara, June 2026.

[2] "Neutrinos: Atmospheric, Solar, Astrophysical, and More." https://npc.fnal.gov/neutrino-university/. Lecture for the NPC Neutrino University at Fermilab, June 2026.

[3] "Neutrino Oscillations and Theory Biases.". Lecture for students at the CETUP workshop in Lead SD, July 2023.

[4] "Neutrino Oscillations and Theory Biases." https://indico.bnl.gov/event/19465/timetable/. Lecture for students at BNL, June 2023.

[5] "Neutrino Oscillations and Theory Biases." https://indico.bnl.gov/event/15829/timetable/. Lecture for students at BNL, June 2022.

[6] "Neutrino Oscillations.". Two lectures for undergraduates at TIFR, May 2021.

Notes

[1] P. B. Denton, H. Minakata, and S. J. Parke, "Comment on 1801.10488v3,". https://zenodo.org/record/1177535.

Code

[1] P. B. Denton and S. J. Parke, "NuFast-Earth.". https://github.com/PeterDenton/NuFast-Earth/.

[2] P. B. Denton and S. J. Parke, "NuFast-LBL.". https://github.com/PeterDenton/NuFast-LBL/.

[3] A. Abdullahi and P. B. Denton, "Astro-Nu-Decay.". https://github.com/PeterDenton/Astro-Nu-Decay.

[4] P. B. Denton, "Peterdenton/nu-pert-compare: v1.0.0," Jan., 2019. https://doi.org/10.5281/zenodo.2547029. https://github.com/PeterDenton/Nu-Pert-Compare.

[5] P. B. Denton, "`ANA` v1.0.0: Astrophysical Neutrino Anisotropy," Mar., 2017. https://doi.org/10.5281/zenodo.438675. https://github.com/PeterDenton/ANA.

[6] P. B. Denton, "`Nu-Pert` v0.2.2: Analytic and compact perturbative expressions for neutrino oscillations in matter," June, 2016. https://doi.org/10.5281/zenodo.54629. https://github.com/PeterDenton/Nu-Pert.

Miscellaneous

[1] P. B. Denton* *et al.*, "Neutrino Non-Standard Interactions." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF3_NF1-CF7_CF0-TF11_ *Editor.

[2] P. B. Denton* and S. J. Parke, "Direct Probes of the Matter Effect in Neutrino Oscillations." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF1_NF3-TF0_TF0_Peter *Editor.

[3] M. Bustamante*, P. B. Denton*, S. Wissel*, *et al.*, "Ultra-High-Energy Neutrinos." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF4_NF6-CF7_CF3-TF9_T *Editor.

[4] P. B. Denton* *et al.*, "Computing Neutrino Oscillations in Matter Efficiently." Snowmass 2021: LOI, July, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF8-CompF2-005.pdf. *Editor.

[5] L. A. Anchordoqui, M. Bustamante, *et al.*, "Cosmic Neutrino Probes of Fundamental Physics." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/CF/SNOWMASS21-CF7_CF1-NF4_NF3-TF11_

[6] L. A. Anchordoqui *et al.*, "Synergy of astro-particle physics and collider physics." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/CF/SNOWMASS21-CF7_CF0-EF6_EF7-NF5_N

[7] D. Soldin *et al.*, "Studies of the Muon Excess in Cosmic Ray Air Showers." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/CF/SNOWMASS21-CF7_CF0-EF6_EF7-AF4_A

[8] J. L. Feng, F. Kling, *et al.*, "Forward Physics Facility." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/EF/SNOWMASS21-EF9_EF6_EF10_EF5-NF6_

[9] L. Johns *et al.*, "Supernova neutrinos and particle-physics opportunities." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF8_NF4-CF3_CF7-TF9_T

[10] K. Scholberg *et al.*, "Neutrino Opportunities at the ORNL Second Target Station." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF6_NF9-CF1_CF0-TF11_

[11] M. Hostert *et al.*, "Opportunities and signatures of non-minimal Heavy Neutral Leptons." Snowmass 2021: LOI, August, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF2_NF3-EF9_EF0-RF4_R

[12] D. A. Sierra *et al.*, "Coherent elastic neutrino-nucleus scattering: Theoretical and experimental impact." Snowmass 2021: LOI, May, 2020. https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF0-002.pdf.

Thesis

[1] P. B. Denton, *Methods for Probing New Physics at High Energies*. PhD thesis, Vanderbilt U., 2016-12-18. https://ir.vanderbilt.edu/handle/1803/12817.